Debayan Das

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Portfolio Github LeetCode Codeforces

Education

Jadavpur University **2023 - 2027**
Bachelor of Technology in Electronics and Telecommunication Engineering *CGPA : 8.43*

Work Experience

AlgoBengal Research Labs **Jun 2026 – Aug 2026**
Cryptography Developer Intern *Kolkata, India*

- **Post-Quantum Signature Flow Separation:** Decomposed a **Digital Signature Scheme(Crystal Dilithium ML-DSA)** into three separate CLI operators - keygen, sign, and verify, using Agentic AI (Claude Code), generated across 3 NIST security levels, reducing manual C refactoring and build-system turnaround from 2 weeks to 2 hours.
- **Multi-Signature Layer:** Delivered 2,400 lines of warning-clean C99 implementing t of n multi-signature scheme over ML-DSA, driving the design specification straight to compiling, benchmarked code via Agentic AI. Measured a 4.02x verification latency reduction at a 100-of-100 quorum.

Projects

Post-Quantum Multi-Signature - ML-DSA/CRYSTALS-Dilithium | *C99, POSIX Threads, Applied Cryptography*

- Built a t-of-n multi-signature layer in C on top of the NIST FIPS 204 (ML-DSA) reference implementation. It has 4 command-line tools (keygen, sign, combine, verify) and supports all 3 security levels (ML-DSA-44/65/87).
- Built multithreaded quorum verification with pthreads and early exit. It cut verification of a 100-signature quorum from 7.66 ms to 1.90 ms (4.0x speedup on 6 cores). Cache-line padding avoids false sharing, threads split work in strided slices to balance load, and I checked that the verify path is safe to run on several threads.

SpeedAI - AI SaaS Platform | *React, Express.js, Node.js, Neon Postgres, Cloudinary, Gemini API* **Live** | **Code**

- Built a full-stack AI SaaS platform putting five distinct AI functionalities together - creating images, erasing backgrounds or unwanted objects from photos, giving resume feedback, writing articles and generating blog titles.
- Built the React frontend with Tailwind CSS and the backend with Node.js and Express, backed by a serverless PostgreSQL database on Neon.
- Secured the API with Clerk authentication and integrated Cloudinary's AI transformations for background and object removal.

High-Performance Cache Simulator | *C++, Multi-threading* **Code**

- Implemented a thread-safe cache engine in C++ with Least Recently Used (LRU) and Least Frequently Used (LFU) eviction policies, using hash maps and doubly linked lists for O(1) get and put operations.
- Implemented multi threading and parallelized the operations across threads for up to 2.7X practical speedup.
- Used std::mutex to ensure safe parallel read/write operations.

Achievements

- **Problem Solving:** Solved **800+ Data Structures & Algorithms problems.**
- **Competitive Programming: LeetCode Knight (2030)** | **Codeforces Specialist (1522)**
- **LeetCode Contest Ranks: Globally #198** in Biweekly Contest 500, **Globally #394** in Weekly Contest 504.
- **Codeforce Contest Ranks: Globally #292** in Codeforces Round 1122(Div3).
- **Siemens Scholar:** Selected as a Siemens Scholar among only 300 candidates in Batch XI across India.

Technical Skills

Languages: C/C++, Python, SQL, JavaScript, Rust, HTML, CSS
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Database Management System, Operating Systems
Backend & Web Node.js, Express, REST APIs, Multi-threading, React, Tailwind CSS
Databases & Cloud: PostgreSQL (Neon serverless), MySQL, Vercel, Cloudinary
AI & Agentic Tools: Claude Code, Gemini API, GitHub Copilot, ChatGPT
Tools: Git, GitHub, Linux, VS Code, Jupyter Notebook.

Leadership

- **Placement Coordinator:** Serving as student placement coordinator in Jadavpur University Training & Placement Cell.
- **Core Organasing Member, F.E.T.S.U. presents Sanskriti'26:** Lead the core organizing committee of F.E.T.S.U. presents Sanskriti'26, delivered 20+ flagship events in 9 days with 35K+ attendees.